# A unified clique-extension lemma: proved research candidate

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This is an independent research candidate, not a modification of the main manuscript or a formal verification. The arguments below are elementary proofs. The separate finite experiment is not used to justify them.

Let $Y$ be a finite graph of order $M \geq 3$, with finite labelled lists $L_f$ for $f \in V(Y)$. Its clique extension $X$ has vertices $(f, c)$, $c \in L_f$; each fibre is complete, and the cross edges are $(f, c)(g, c)$ for $fg \in E(Y)$ and $c \in L_f \cap L_g$. Assume $|L_f| \geq 3$ and $|L_f \cap L_g| \geq 2$ on every base edge. For distinct labels $p, q$, let $B_{pq}$ be the spanning subgraph of $Y$ whose edges are exactly those with $L_f \cap L_g = \{p, q\}$.

**Lemma 1** (Exact obstruction for cyclic bridge selection)**.** *Every base cycle admits bridge labels from its successive intersections, with consecutive bridge labels distinct, if and only if every $B_{pq}$ is bipartite.*

*Proof.* Lists of size at least two on a cycle admit a proper list colouring unless the cycle is odd and every list is the same pair. For completeness, a list of size at least three can be left until last; otherwise unequal consecutive binary lists permit a greedy traversal beginning with a label absent from the last list. Equal binary lists are coloured by alternation exactly on even cycles. Thus a failing cycle is precisely an odd cycle of some $B_{pq}$. □

**Theorem 2** (Three clique-extension conclusions)**.** *Suppose every $B_{pq}$ is bipartite. Consider also the following conditions:*

*(E) For every $f$ and $c \in L_f$, some neighbour $g$ of $f$ has $c \in L_g$.*

*(R) Every base path containing all but at most one vertex has an edge whose intersection list has size at least three.*

*Then:*

1. *if $Y$ is Hamiltonian, $X$ is Hamiltonian;*
2. *if $Y$ is vertex-pancyclic and (E) holds, $X$ is vertex-pancyclic;*
3. *if $Y$ is Hamilton-connected and (E),(R) hold, $X$ is Hamilton-connected. This third conclusion does not require the bipartiteness assumptions.*

*Proof.* First, a path whose intersection lists have size at least two, with one of size at least three, lifts with arbitrary prescribed endpoint labels. Choose bridge labels greedily from the two ends towards the three-element list, choosing there last. This avoids each prescribed endpoint and makes successive bridge labels distinct. Spanning paths in the complete fibres then concatenate.

For a cycle $f_0, \ldots, f_{s-1}, f_0$, choose cyclic bridge labels by the lemma. Spanning fibre paths give a spanning cycle on all its fibres, proving conclusion 1 for a Hamilton base cycle. More generally, given $x = (f_0, a)$, choose the path in its fibre on between 3 and $|L_{f_0}|$ vertices, containing $x$, and the other fibre paths on between 2 and $|L_{f_i}|$ vertices. This gives every length in

$$[2s+1,\ \sum_{i=0}^{s-1} |L_{f_i}|]$$

through $x$. Condition (E) gives an edge $f_0 g$ preserving $a$ and another common label $p \neq a$. The two cross edges at $a, p$, with fibre paths on two or three vertices, give lengths $4, 5, 6$ through $x$; its own fibre gives length 3. For each $3 \leq s \leq M$, take a base cycle of length $s$ through $f_0$, and denote its total fibre size by $T_s$. The intervals $[2s+1, T_s]$ overlap since $T_s \geq 3s$ and $2s+3 \leq 3s$; moreover $T_M = |V(X)|$. This proves conclusion 2.

For conclusion 3, endpoints in different fibres are handled by a base Hamilton path and (R). For endpoints $(f, a), (f, b)$ in one fibre, choose $c \in L_f \setminus \{a, b\}$ and a neighbour $g$ preserving $c$ by (E). Every edge of the Hamilton-connected base lies on a Hamilton cycle $C$. Let $h$ be the other neighbour of $f$ on $C$ and choose $q \in (L_f \cap L_h) \setminus \{c\}$. Interchange $a, b$ if $q = a$, reversing the final path. Now split the root fibre into paths $(f, a) \rightsquigarrow (f, c)$ and $(f, q) \rightsquigarrow (f, b)$; use a singleton for the second when $q = b$, otherwise its endpoint edge, and put every remaining vertex in the first. Condition (R) lifts $C - f$ through every other fibre from $(g, c)$ to $(h, q)$. Join these three paths by the cross edges at $c, q$. □

**Lemma 3** (Recolouring interfaces have the required bipartition)**.** *Let $H = G - v \neq \varnothing$ and $k \geq \deg_G(v) + 3$. With $Y = \mathcal{R}_k(H)$ and $L_f = [k] \setminus f(N_G(v))$, every $B_{pq}$ is bipartite.*

*Proof.* At an edge of $B_{pq}$, the one-coordinate change implies $2 \geq |L_f| - 1$, so both endpoint lists have size three. Hence $\deg_G(v) = k - 3$, and its neighbours use pairwise distinct colours, none equal to $p, q$. Order these neighbours. Append to their colour tuple the unique unused colour in $[k] \setminus \{p, q\}$. This is a permutation, and a $B_{pq}$ edge swaps one neighbour entry with the last entry. Permutation sign gives a bipartition. Isolated vertices of $B_{pq}$ may be assigned arbitrarily. □

**Corollary 4** (Application limits)**.** *The preceding lemmas give one-vertex Hamiltonian lifting without any global colour-permutation argument. They give vertex-pancyclic lifting whenever (E) is available. They give Hamilton-connected lifting when also $v$ has a non-neighbour and (E) is available.*

*Proof.* A non-neighbour $w$ supplies (R): colour permutations give all $k \geq 3$ labels at $w$, so any path omitting at most one base vertex must change $w$, preserving the full available list. Condition (E) follows either from $\delta(Y) > \deg_G(v)$ (at most $\deg_G(v)$ moves can destroy a fixed available label), or from $H$ being nonempty and $d$-degenerate with $k \geq d + 3$ (recolour a low-degree vertex while avoiding its current label and the fixed label). □

**What remains.** This unifies the existing lifting mechanism and isolates its exact cyclic obstruction. It does not remove the complete-graph branch of degenerate Hamilton-connectedness, because that branch has no non-neighbour to guarantee (R). It does not establish that (R) may be deleted. No new palette threshold is claimed.